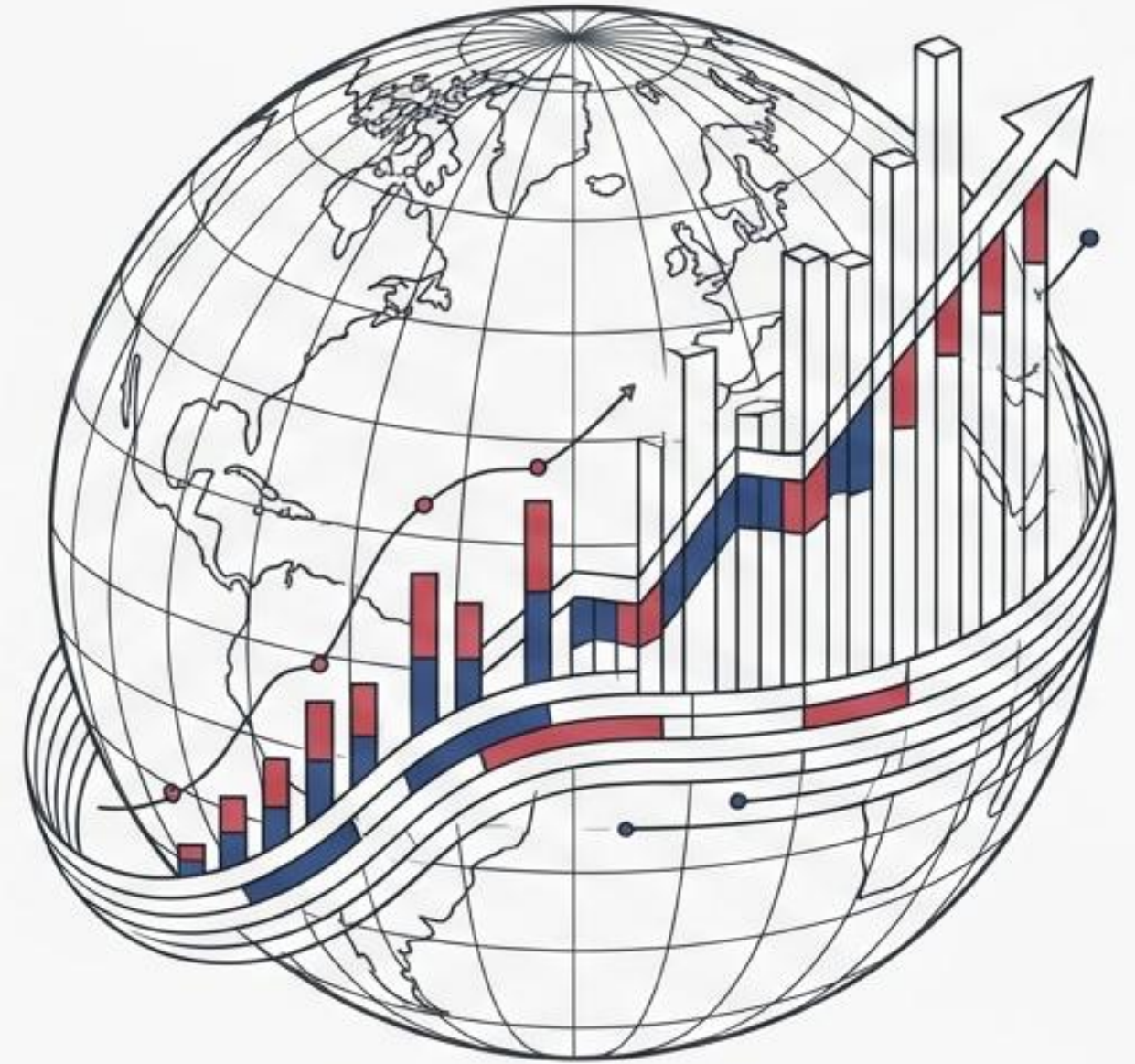


GLOBAL VIDEO GAME MARKET INTELLIGENCE

Exploratory Data Analysis & Strategic Insights



AUTHOR	:: Atharv Milind Suryavanshi
COURSE	:: ITRS010 – Exploratory Data Analysis
RECORDS	:: 16,598 Global Titles
SOURCE	:: Kaggle Datasets

Project Parameters and Analytical Vectors

THE OBJECTIVE

Transform raw Kaggle dataset records into actionable market intelligence by identifying historical sales trends, platform dominance, and publisher performance.

THE TOOLKIT



Python



Pandas



Matplotlib



Jupyter

INVESTIGATION ZONES

1

Macro Yearly Trends



2

Publisher & Platform Dynamics



3

Consumer Genre Preferences



4

Individual Product Power Laws

Top 10 Rows in Given Data Set Uncleaned



df

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
0	1	Wii Sports	Wii	2006.0	Sports	Nintendo	41.49	29.02	3.77	8.46	82.74
1	2	Super Mario Bros.	NES	1985.0	Platform	Nintendo	29.08	3.58	6.81	0.77	40.24
2	3	Mario Kart Wii	Wii	2008.0	Racing	Nintendo	15.85	12.88	3.79	3.31	35.82
3	4	Wii Sports Resort	Wii	2009.0	Sports	Nintendo	15.75	11.01	3.28	2.96	33.00
4	5	Pokemon Red/Pokemon Blue	GB	1996.0	Role-Playing	Nintendo	11.27	8.89	10.22	1.00	31.37
...	...	...	...	...	...	...	...	...	...	...	...
16593	16596	Woody Woodpecker in Crazy Castle 5	GBA	2002.0	Platform	Kemco	0.01	0.00	0.00	0.00	0.01
16594	16597	Men in Black II: Alien Escape	GC	2003.0	Shooter	Infogrames	0.01	0.00	0.00	0.00	0.01
16595	16598	SCORE International Baja 1000: The Official Game	PS2	2008.0	Racing	Activision	0.00	0.00	0.00	0.00	0.01
16596	16599	Know How 2	DS	2010.0	Puzzle	7G//AMES	0.00	0.01	0.00	0.00	0.01
16597	16600	Spirits & Spells	GBA	2003.0	Platform	Wanadoo	0.01	0.00	0.00	0.00	0.01

16598 rows × 11 columns



df.isnull().sum()

	0
Rank	0
Name	0
Platform	0
Year	271
Genre	0
Publisher	58
NA_Sales	0
EU_Sales	0
JP_Sales	0
Other_Sales	0
Global_Sales	0

dtype: int64

df.info()

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RangeIndex: 16598 entries, 0 to 16597

Data columns (total 11 columns):

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1	Name	16598 non-null	object
2	Platform	16598 non-null	object
3	Year	16327 non-null	float64
4	Genre	16598 non-null	object
5	Publisher	16540 non-null	object
6	NA_Sales	16598 non-null	float64
7	EU_Sales	16598 non-null	float64
8	JP_Sales	16598 non-null	float64
9	Other_Sales	16598 non-null	float64
10	Global_Sales	16598 non-null	float64

dtypes: float64(6), int64(1), object(4)

memory usage: 1.4+ MB

Dataset After Cleaning



df

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
0	1	Wii Sports	Wii	2006.0	Sports	Nintendo	41.49	29.02	3.77	8.46	82.74
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...	...	...	...	...	...	...	...	...	...	...	...
16593	16596	Woody Woodpecker in Crazy Castle 5	GBA	2002.0	Platform	Kemco	0.01	0.00	0.00	0.00	0.01
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16596	16599	Know How 2	DS	2010.0	Puzzle	7G//AMES	0.00	0.01	0.00	0.00	0.01
16597	16600	Spirits & Spells	GBA	2003.0	Platform	Wanadoo	0.01	0.00	0.00	0.00	0.01

16540 rows × 11 columns



df.isnull().sum()

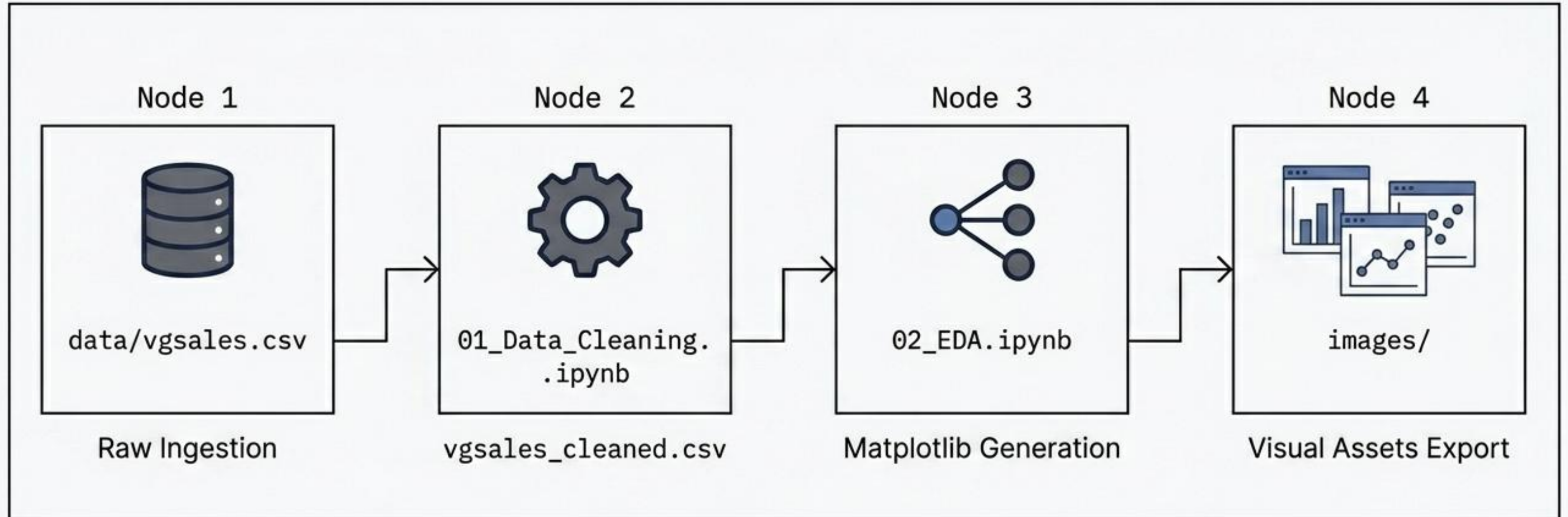
	0
Rank	0
Name	0
Platform	0
Year	0
Genre	0
Publisher	0
NA_Sales	0
EU_Sales	0
JP_Sales	0
Other_Sales	0
Global_Sales	0

dtype: int64

df.info()

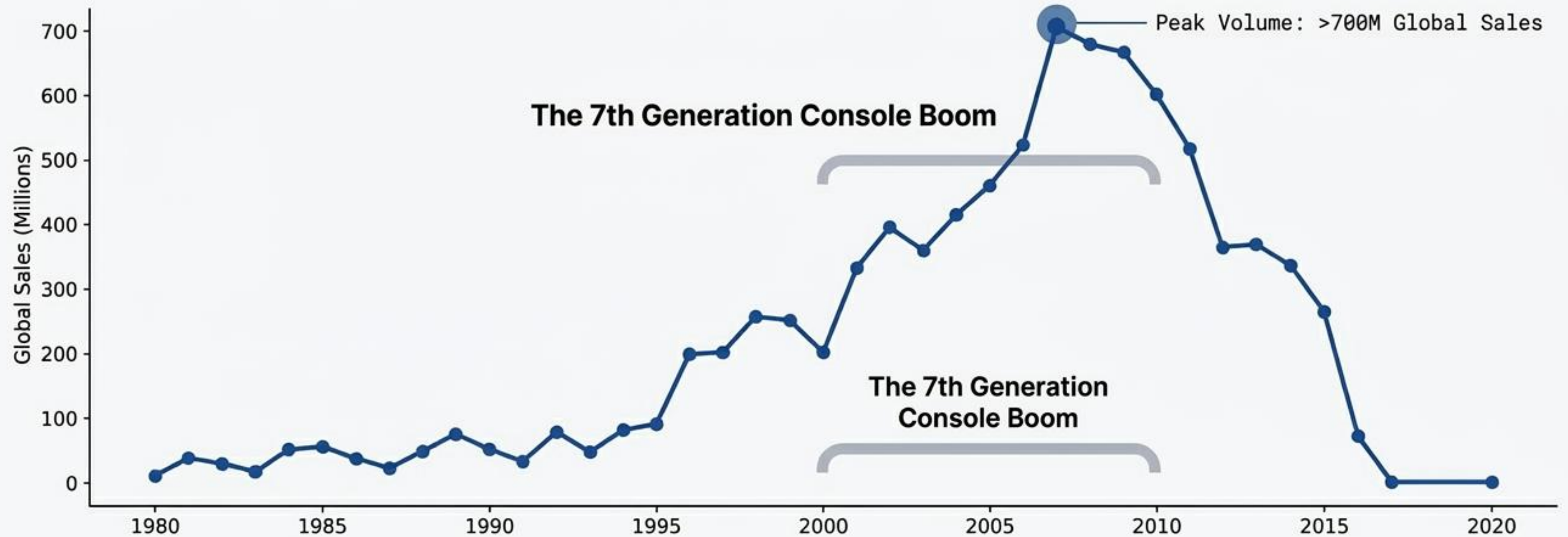
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Data columns (total 11 columns):
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1   Name            16540 non-null  string
2   Platform        16540 non-null  string
3   Year            16540 non-null  float64
4   Genre           16540 non-null  string
5   Publisher       16540 non-null  string
6   NA_Sales        16540 non-null  float64
7   EU_Sales        16540 non-null  float64
8   JP_Sales        16540 non-null  float64
9   Other_Sales     16540 non-null  float64
10  Global_Sales    16540 non-null  float64
dtypes: float64(6), int64(1), string(4)
memory usage: 1.5 MB
```


Data Architecture and Processing Pipeline



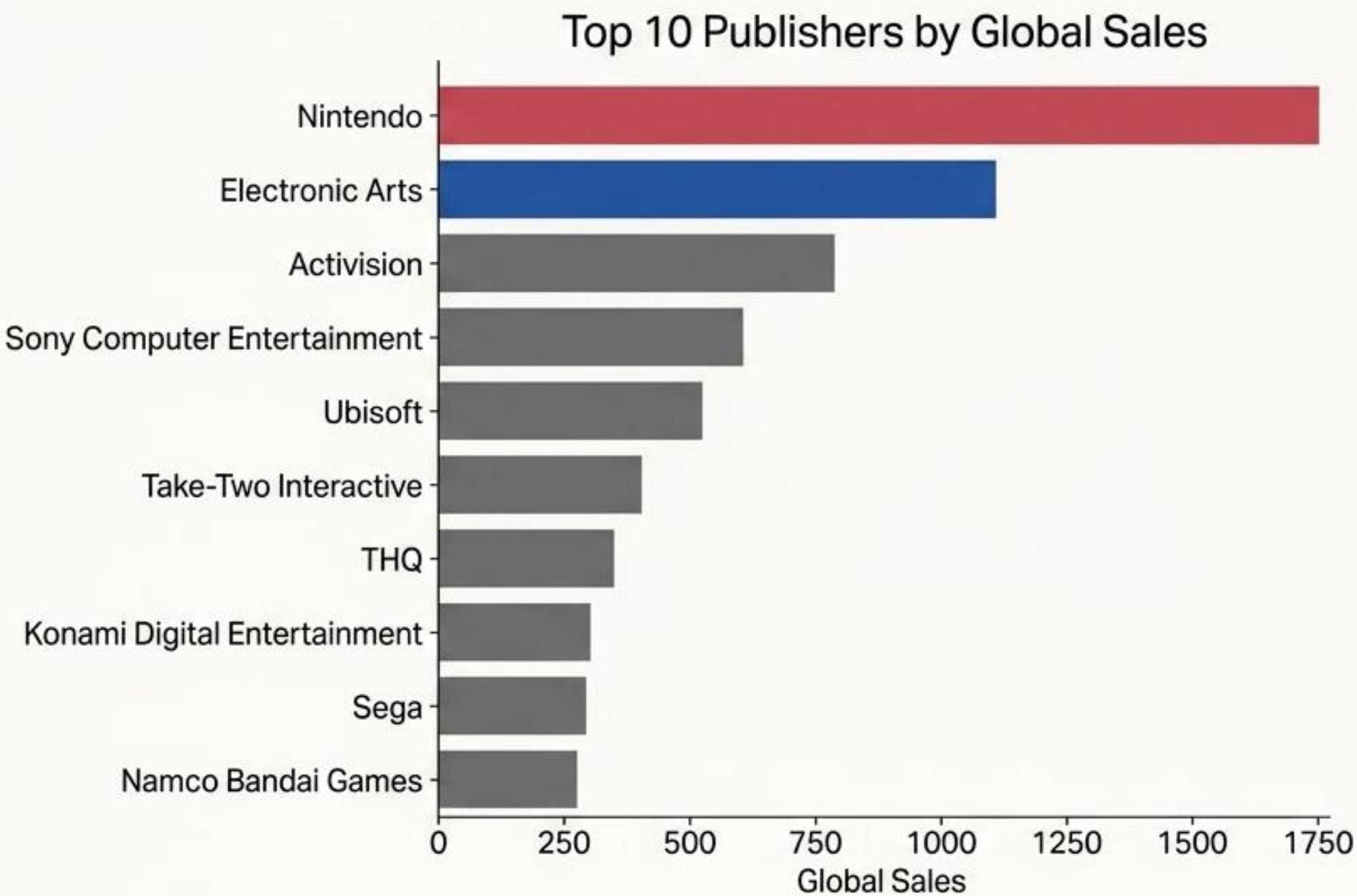
Pipeline executes sequential transformation from unfiltered Kaggle records into structured, high-fidelity analytical assets.

The 2000s Market Boom and Correction



Global sales experienced unprecedented, explosive growth throughout the 2000s, driven by mass-market console adoption, before facing a steep market correction in the subsequent decade.

Publisher Dominance and Strategic Divergence



Nintendo (Rank #1)

Total Vol: ~1750M

The Blockbuster Engine

Reliance on first-party, hardware-moving exclusive franchises.

Electronic Arts (Rank #2)

Total Vol: ~1100M

The Annualized Volume

Reliance on consistent, multi-platform sports and action releases.

Nintendo outpaces the runner-up by over 600M global sales, establishing unquestioned market leadership.

The Hardware Triumvirate

PlayStation 2



MARKET SHARE: ~45%

Manufacturer: Sony

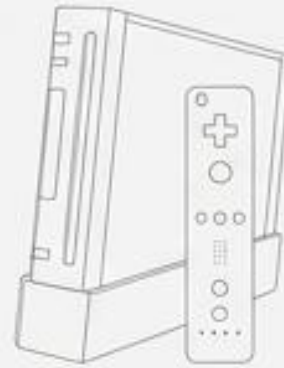
Launch: 2000

Units: >155M

Media: DVD

Key Feature: Backward Compatibility

Wii



MARKET SHARE: ~32%

Manufacturer: Nintendo

Launch: 2006

Units: ~101M

Media: Wii Optical Disc

Key Feature: Motion Control, Broad Appeal

Xbox 360



MARKET SHARE: ~23%

Manufacturer: Microsoft

Launch: 2005

Units: ~84M

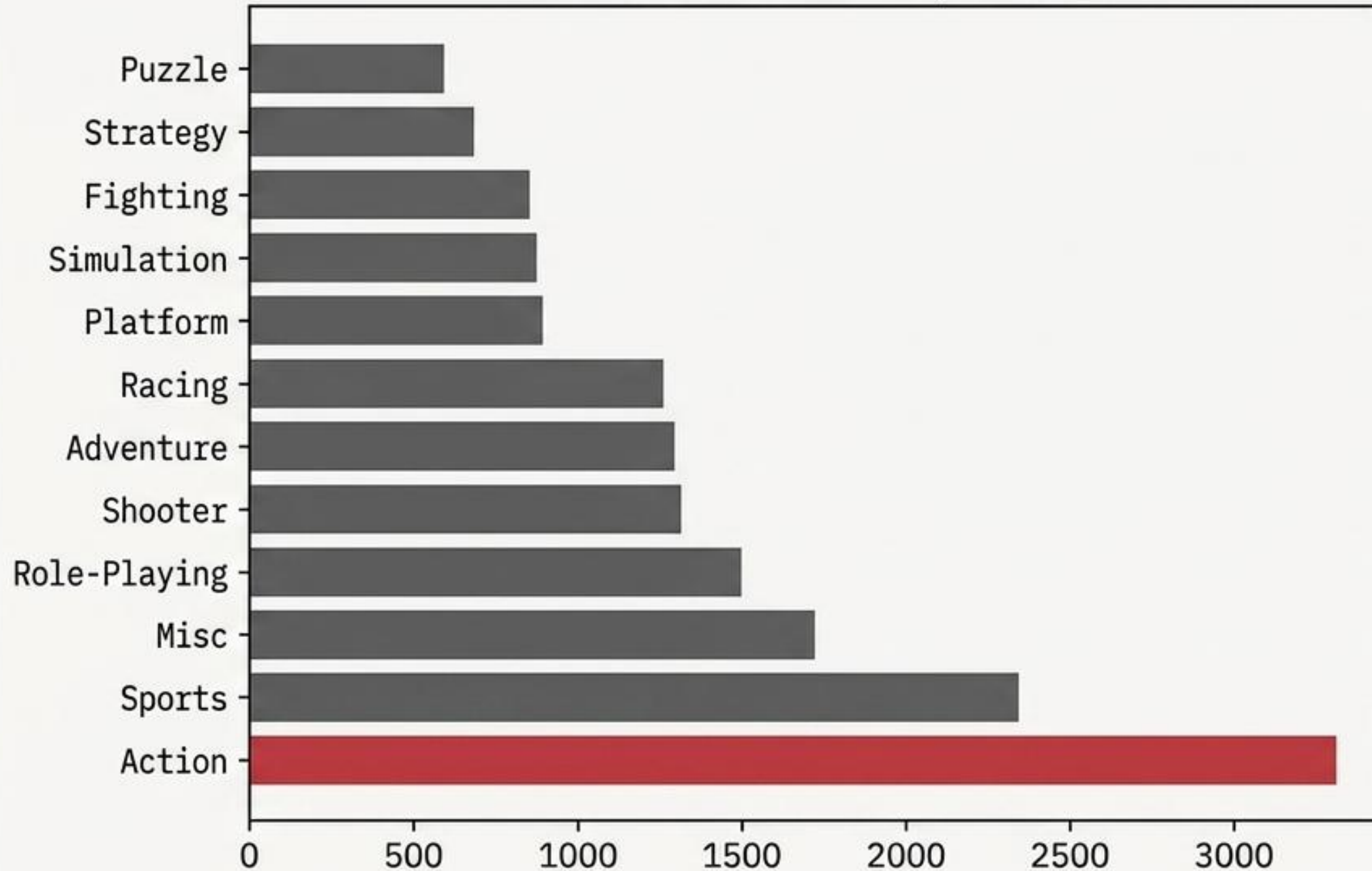
Media: DVD/HD-DVD (optional)

Key Feature: Online Gaming (Xbox Live), Multimedia

The historical market peak was not platform-monopolized. It was sustained by three distinct hardware titans—PS2, Wii, and Xbox 360—each capturing massive, distinct consumer demographics.

The Action Genre Production Paradox

Number of Games By Genre



Production Telemetry

Action :: ~3,300 titles

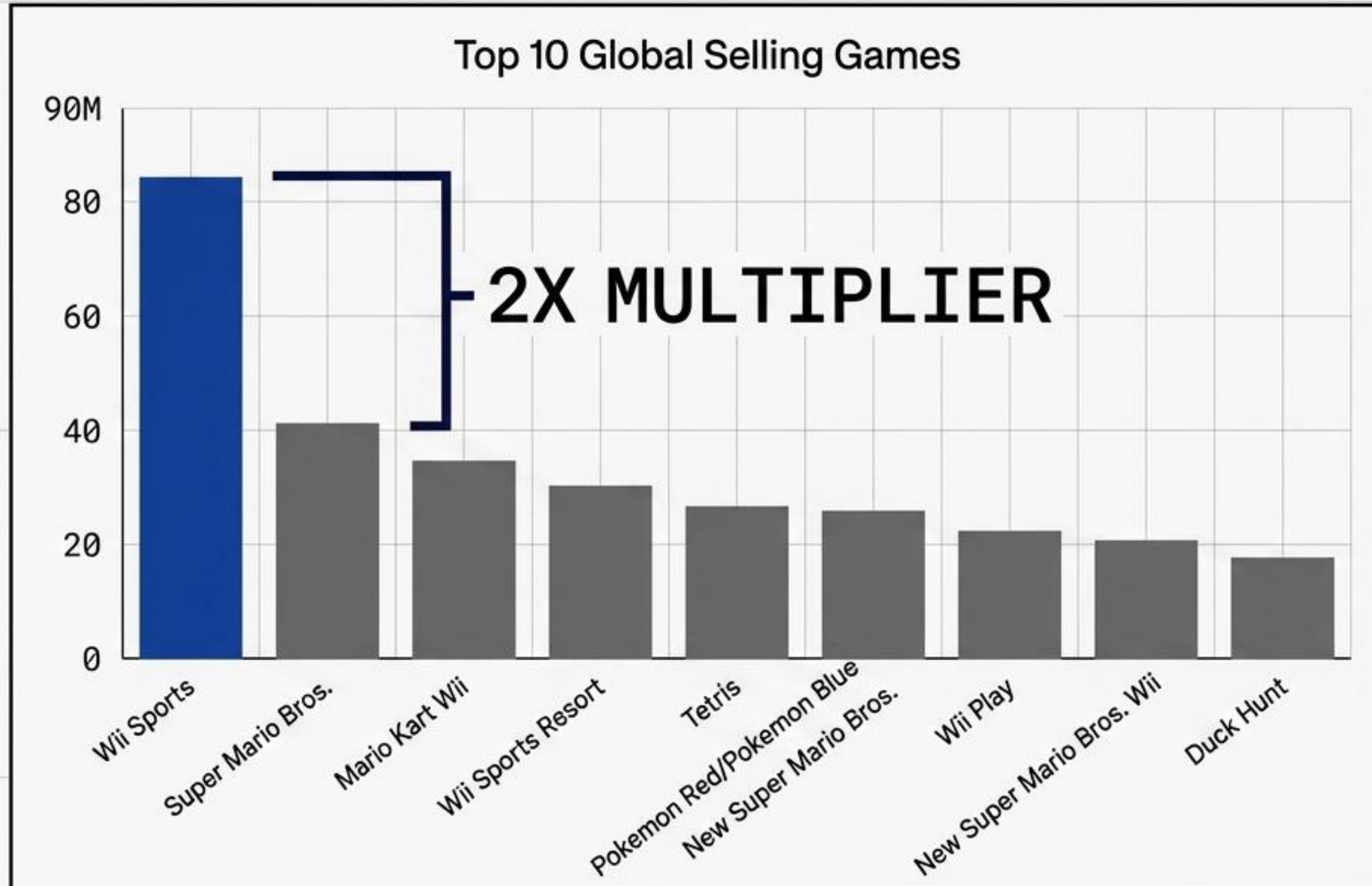
Sports :: ~2,300 titles

Misc :: ~1,700 titles

Strategic Insight

While Action is identified as one of the most popular genres, the sheer volume of titles produced indicates a highly contested, oversaturated market space for developers compared to niche genres.

The Blockbuster Power Law



A fraction of titles generate the majority of global revenue.

Wii Sports did not just win the market; it sold more than double the volume of the next highest-selling game in history.

Source: Global Sales Data, Internal Analysis

Unified Theory of the Market Peak

Macro Trend: Global market mathematically peaks in the late 2000s.

Dominance: Nintendo holds the #1 publisher position by a >600M unit margin.

The historic peak of the global video game market was engineered by a singular strategy: Nintendo's expansion into the casual, family-gaming demographic via the Wii ecosystem.

Power Law: Wii Sports and Mario Kart Wii shatter all individual product sales records.

Market Reality Readout

Blockbuster Reliance

The industry is structurally dependent on massive outlier hits rather than median success.

Nintendo's Blueprint

First-party, family-friendly IP generates unmatched lifetime sales volume.

Action Oversaturation

High consumer popularity is significantly offset by severe market crowding.

The Hardware Cycle

Unprecedented industry growth is directly tied to disruptive hardware cycles (e.g., the 7th Gen).

Thank You